



RISING WATERS

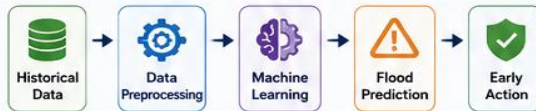
FLOOD PREDICTION SYSTEM

AI-POWERED EARLY WARNING FOR A SAFER TOMORROW

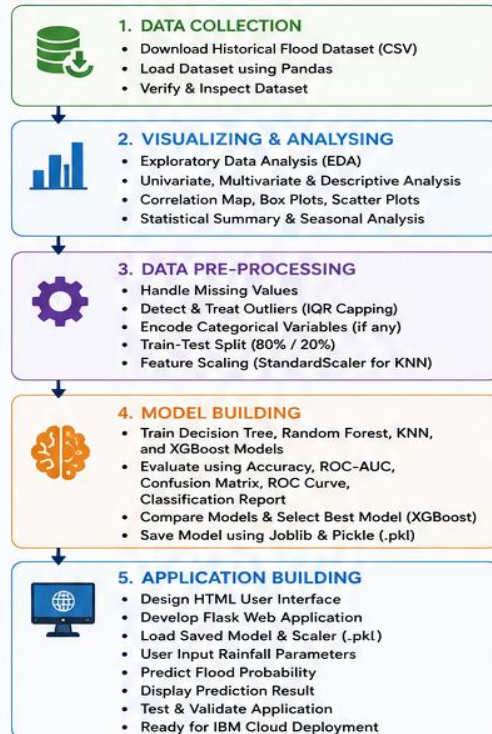


PROJECT OVERVIEW

The Rising Waters – Flood Prediction System uses Machine Learning to analyze historical rainfall data and seasonal weather patterns to predict flood risk accurately. The system helps authorities and communities make timely decisions and reduce the impact of flood-related disasters.



PROJECT WORKFLOW



MACHINE LEARNING MODELS



MODEL PERFORMANCE COMPARISON

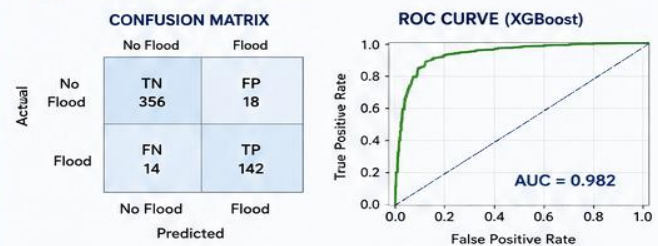
Model	Accuracy	ROC-AUC Score
Decision Tree	85.21%	0.883
Random Forest	92.31%	0.948
K-Nearest Neighbors (KNN)	89.14%	0.912
XGBoost (Best Model)	96.55%	0.982

BEST MODEL

Accuracy
96.55%



MODEL EVALUATION



CLASSIFICATION REPORT (XGBoost)

	Precision	Recall	F1-Score	Support
No Flood	0.96	0.95	0.95	374
Flood	0.89	0.91	0.90	156
Accuracy			0.9655	530
Macro Avg	0.93	0.93	0.93	530
Weighted Avg	0.96	0.97	0.96	530

KEY METRICS (XGBoost)

Accuracy	96.55%
ROC-AUC Score	0.982
Precision	0.96
Recall	0.97
F1-Score	0.96



WEB APPLICATION



PROJECT ACHIEVEMENTS

- ✓ Collected and analyzed historical flood data
- ✓ Performed comprehensive EDA
- ✓ Applied complete data preprocessing techniques
- ✓ Developed and compared 4 ML classification models
- ✓ Selected XGBoost with 96.55% accuracy
- ✓ Saved the model using Joblib & Pickle
- ✓ Designed and developed Flask Web Application
- ✓ Integrated ML model with web interface
- ✓ Built a scalable solution for IBM Cloud deployment



REAL-WORLD IMPACT

-  Government & Disaster Management Agencies
-  Weather Forecasting Organizations
-  Emergency Response Teams
-  Smart City Infrastructure Planners
-  Water Resource Management Authorities
-  Researchers in Climate Resilience



FUTURE SCOPE

-  Real-time Weather APIs
-  IoT & Sensor Integration
-  Satellite & GIS Visualization
-  SMS & Email Alerts
-  Mobile Application Support
-  IBM Cloud Deployment
-  Deep Learning & Time-Series Models



RISING WATERS

Combining Data, AI, and Technology for a Safer and Flood-Resilient Future.

